Karol Chojnacki

Quantitative Researcher with ML MSc and Math BSc from MIMUW.
Experienced in high-frequency orderbook analysis, PyTorch framework design, and algorithmic trading.

Contact

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Experience

Quantitative Researcher @ IDEAS Research Institute *Feb 2026 – Present*

- L2 Orderbook Δ analysis on crypto – alpha generation, market impact, slippage.
- Designed and implemented a modular PyTorch-based training framework for ML experimentation and backtesting (*OHLCV-1min, L2 orderbook-100ms*).
- Researching robustness and sparsity in models via optimizers.

Algorithmic Quant - Power Trading @ Respect Energy *Aug 2023 – May 2024*

- Developed, backtested and traded with statistical arbitrage models in illiquid XBID Intraday energy markets.
- Analyzed orderbook L3 Δ's dynamics to identify intraday microstructure patterns.

Math Olympiad Tutor *Aug 2020 – Sept 2022*

- Ran weekly problem-solving sessions preparing for Polish Mathematical Olympiad. Focused on inequalities, mathematical analysis and algebra.

Education

MSc Machine Learning Oct 2024 – Aug 2026
University of Warsaw *Warsaw, Poland*

- Thesis: *Adversarial Robustness of Optimizers in Medical Imaging*

BSc Mathematics, BSc Economics Oct 2020 – July 2024
University of Warsaw *Warsaw, Poland*

- Math Thesis: *Deep Q-Learning in Algorithmic Trading*.
- Econ Thesis: *Ensembled LSTM with Walk Forward Optimization in Algorithmic Trading*.

Interesting projects

- **TCL Research Europe - Image Quality Assessment**
  Developing Vision Language Models for IQA problems using PyTorch (proprietary research).
- **Explainable BiomedCLIP**
  This repository provides tools for finetuning BiomedCLIP on medical imaging datasets and explaining its predictions using multiple explainability methods.
- **TradingBot**
  Designed modular backtesting framework for equity trading and model tuning for price prediction.

Grants and achievements

- **Ministry of Education's "School of Eagles" Research Grant**
  Developed an ensemble of models to enhance time-series prediction on S&P500.
- University of Warsaw **Integration Bee** 2025 1st, 2026 2nd place.

Selected Courses

- **MSc ML**: Reinforcement Learning, Computer Vision, Natural Language Processing, Statistical ML, Statistical Data Analysis, Bayesian Statistics, Graphical Models.
- **BSc Mathematics**: Probability, Statistics, ODEs, Discrete Mathematics, Linear Algebra.

IT Skills

- **Programming languages:** Python.
- **Libraries:** PyTorch, NumPy, Pandas, scikit-learn.
- **Tools:** Git, ClearML, Linux, Bash, Slurm, LaTeX, QuestDB (SQL).

Spoken languages

Polish (native), English (C1).

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